CncCommunication.dll

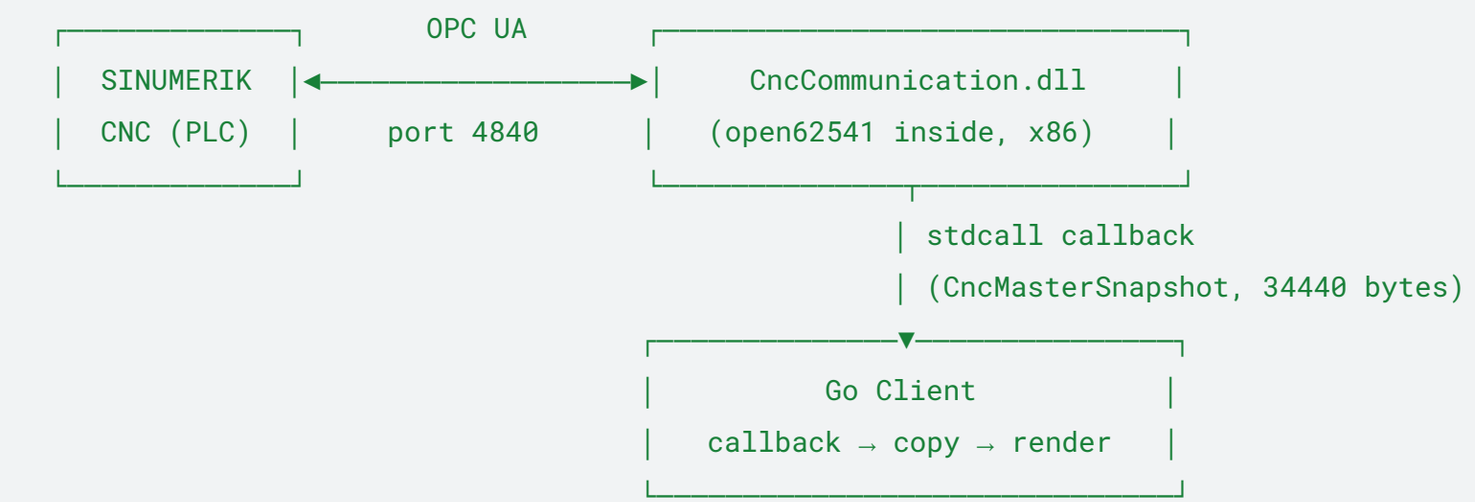
1. Overview

CncCommunication.dll is a dynamic link library implemented in C++ designed to stream real-time telemetry from industrial Computer Numerical Control (CNC) controllers directly to external software clients via asynchronous event loops.

The Go application interfaces with the DLL by loading exports, issuing a connection call (CNCConnect), and handling inbound updates through unmanaged callback pointers dispatching instances of CncMasterSnapshot.

The DLL encapsulates the **entire OPC UA session** — endpoint discovery, secure channel negotiation, and certificate handling — so the Go client requires **no OPC UA certificate bundles, zip archives, or session configuration**. It only supplies the controller address, port, and credentials.

2. System Architecture



Data flow (event-driven, no polling in the client):

1. The Go client loads the DLL and registers two stdcall callbacks (syscall.NewCallback).
2. CNCConnect starts a background worker thread inside the DLL.
3. The worker polls the controller, assembles a CncMasterSnapshot, and invokes the data callback with a pointer to the buffer.
4. The callback **copies** the payload into the managed Go struct and signals the renderer via a buffered channel.
5. The main goroutine renders the live ANSI dashboard on each signal.

3. Package Artifacts & Dependencies

Filename	Dependency Domain	Description
open62541.dll	OPC UA Runtime	Open-source C-based OPC UA stack.
libcrypto-3.dll, libssl-3.dll	OpenSSL	TLS encryption and handshake providers for OPC UA transport.
Fwlib32.dll, fwlibe1.dll	Fanuc FOCAS2	Low-level proprietary protocol drivers for Fanuc controllers.
CncCommunication.dll	Core Runtime	Primary orchestration and export library.
client_key.der (.pem)	Security Profile	Private key corresponding to client certificate.
client_cert.der (.pem)	Security Profile	X.509 client certificate for Siemens OPC UA authentication.

3.1 Environment & Prerequisites

Requirement	Value
Target Architecture	**32-bit DLL (x86)** — compile the Go client with set GOARCH=386
Credentials	SIEMENS_USER / SIEMENS_PASS environment variables set before CNCConnect

4. API Reference

4.1 Function Signatures

```C/C++
   // Starts the background worker, opens network sockets, and begins polling
   int32_t CNCConnect(
       const char* ip,          // controller IP address (ANSI, NUL-terminated)
       int32_t     port,        // controller port (e.g. 4840)
       uintptr_t   dataCb,      // stdcall callback: void OnData(const CncMasterSnapshot*)
       uintptr_t   errCb,       // stdcall callback: void OnErr(int32_t code, const char* msg)
       int32_t     machineType  // engine id, see §4.2
   );

   // Stops polling threads and releases underlying protocol sockets
   void CNCdisconnect(int32_t machineId);
```

4.2 Machine Identifier Mapping

Enum Value	Machine Controller Type	Driver Protocol
0	Simulator	Local loopback mock engine
1	Fanuc	FOCAS2 Ethernet (fwlib32)
2	**Siemens**	**Sinumerik OPC UA *(active profile in this project)***
3	Syntec	Syntec Open API / TCP
4	Mitsubishi	

5. DLL Loading & Binding (Go)

```Go
package main

import (
    "fmt"
    "os"
    "syscall"
)

var (
    mod            = syscall.NewLazyDLL("CncCommunication.dll")
    procConnect    = mod.NewProc("CNCConnect")
    procDisconnect = mod.NewProc("CNCdisconnect")
)

func init() {
    // Eagerly resolve the DLL handle at application startup
    if err := mod.Load(); err != nil {
        fmt.Fprintf(os.Stderr, "FATAL: Failed loading CncCommunication.dll: %v\n", err)
        os.Exit(1)
    }
}
```

6. Snapshot Data Structure

6.1 Layout

CncMasterSnapshot matches the C++ definition 1:1 (#pragma pack(8), x86) and totals exactly **34,440 bytes**. The Go client asserts this at startup:

```Go
const expectSize = 34440
if unsafe.Sizeof(CncMasterSnapshot{}) != expectSize {
    // abort — struct drift between client and DLL
}
```

6.2 Data Field Reference

Struct Member	Type	Units / Encoding	Description
IsConnected	int16	Boolean (0/1)	Network connection state
ResultStatus	int16	Error Flag	Controller operational status register
SampleTimestampMs	int64	Milliseconds	Epoch timestamp of snapshot emission
MachineMode	int16	Enum (0=JOG, 1=MDA, 2=AUTO)	Controller operating mode
ProcStatus	int16	Enum (0=Reset, 1=Stop, 2=Run)	NC channel execution state
IsEmergency	int16	Boolean (0/1)	Emergency stop (E-STOP) state
AxisNames	[5][4]byte	ASCII Strings	Axis designations (e.g., X, Y,

Struct Member	Type	Units / Encoding	Description
			Z, B, C)
MachPos	[5]float64	Millimeters / Degrees	Machine Coordinate System (MCS) positions
AbsPos	[5]float64	Inches / Degrees	Work Coordinate System (WCS) positions
RelPos	[5]float64	Inches / Degrees	Relative incremental offsets
DistToGo	[5]float64	Inches / Degrees	Remaining vector to destination
SpindeOverrideActual	float64	RPM	Active spindle surface velocity
SpindleLoadPercetnage	int16	Percent (%)	Spindle drive current load
SpindeOverridePercentage	int16	Percent (%)	Spindle potentiometer position
FeedRateActual	float64	mm/min or in/min	Active linear tool translation rate
FeedRatePercetnage	int16	Percent (%)	Feedrate override potentiometer percentage
AxisLoadX/Y/ZPercentage	int16	Percent (%)	Servo axis drive loads
ProgNumber	int32	Integer ID	Program sequence identifier (O-number)
CurProgLine	int32	Line Count	Current block executing in interpreter
TotalProgLines	int32	Line Count	Total line count of active file
ProgName	[64]byte	ASCII String	File name of active NC program
ActiveProgPath	[256]byte	ASCII Path	Storage path of active NC artifact

Struct Member	Type	Units / Encoding	Description
ToolNumber	int32	Slot/ID	Spindle-mounted active tool identifier
ToolLength	float64	Metric / Imperial	Primary height compensation value
ToolRadius	float64	Metric / Imperial	Cutter radius compensation value
ActiveGCode	[16]byte	ASCII	Currently active G-code group
GcodeGroup	[512]byte	ASCII (reused)	DLL reuses this field for the ";"-separated missing-node list
AlarmNo	int32	Code (0 = OK)	Primary alarm or fault number
AlarmMsg	[128]byte	UTF-8 / ASCII	Alarm description message

7. Asynchronous Callback Architecture

Because callbacks originate from an unmanaged C++ worker thread, callbacks must remain **strictly non-blocking**.

Go

```go
package main

import (
    "fmt"
    "sync"
    "syscall"
    "unsafe"
)

var (
    dataMutex  sync.RWMutex
    lastRecord CncMasterSnapshot
    hasRecord  bool

    // Declare callbacks as package-level singletons
    dataCallback = syscall.NewCallback(onDataCallback)
    errCallback  = syscall.NewCallback(onErrorCallback)
)

func onDataCallback(ptr uintptr) uintptr {
    if ptr == 0 {
        return 0
    }
    // Copy unmanaged memory buffer into managed Go struct
    dataMutex.Lock()
    lastRecord = *(*CncMasterSnapshot)(unsafe.Pointer(ptr))
    hasRecord  = true
    dataMutex.Unlock()
    return 0
```

```
}

func onErrorCallback(code, messagePtr uintptr) uintptr {
    fmt.Printf("[DLL ERROR] Event Code: %d\n", code)
    return 0
}
```

8. Running the Go Client

```shell
cd C:\Avim\Prinsi\G0_V0.0.2_DLL
set GOARCH=386
go run sinumerik_client.go            # live dashboard
go run sinumerik_client.go check      # OPC UA node existence report
```

Press **Ctrl+C** in the dashboard to disconnect cleanly.

Check-mode OPC UA node mapping

Data	OPC UA Node Path
Operating mode	/Bag/State/opMode[u1]
Program status	/Channel/State/acProg[u1,1]
Spindle speed / override	/Channel/Spindle/actSpeed[u1,1], speedOvr[u1,1]
Feed rate / override	/Channel/State/actFeedRateIpo[u1], feedRateOvr[u1,1]
Axis / spindle loads	/Channel/MachineAxis/aaLoad[u1,1..4]
Positions	/Channel/MachineAxis/actToolBasePos[u1,n], /Channel/GeometricAxis/aaPcsRel[u1,n]
Program info	/Channel/ProgramInfo/progName[u1,1], actLineNumber
Tool	/Channel/State/actTNumber[u1,1]

9. Controller-Specific Profiles

9.1 Siemens Sinumerik (OPC UA) — Engine ID: 2

- **Default Ports:** 4840 (Native NCU) or 4841 (custom industrial port profile)
- **Security Certificates:** place client_cert.der and client_key.der directly in the active binary directory.
- **Trust Requirements:** the client certificate must be signed or marked as trusted within the Sinumerik **NCU Access Administration** interface. Connections using unapproved certificates are rejected with status BadRequestNotAllowed.
- **Credentials:** passed to the DLL via SIEMENS_USER / SIEMENS_PASS environment variables, set **before** calling CNCConnect.

9.2 Fanuc — Engine ID: 1

- **Default Port:** 8193 (default FOCAS interface)
- **Dependencies:** requires Fwlib32.dll and fwlibe1.dll present in system lookup paths.

9.3 Syntec — Engine ID: 3

- **Default Port:** typically 5001 or controller-assigned port.

10. Diagnostics & Troubleshooting

Observable Error / Symptom	Root Cause Analysis	Remediation Step
0xC0000135 / DLL load failed: module could not be found	Missing transitive native dependencies — a dependency DLL is not resolvable.	All 6 DLLs must be in Lib\ next to the client. Run dumpbin /dependents CncCommunication.dll to find and add missing dependent .dll files.
%1 is not a valid Win32 application	32-bit DLL loaded by a 64-bit host process.	Compile the Go client with set GOARCH=386; validate the GOARCH setting against the DLL architecture.
Memory Panic / Corrupted floats	Struct size mismatch or memory alignment issues.	Verify unsafe.Sizeof(CncMasterSnapshot{}) == 34440. Validate GOARCH setting against DLL architecture.
CNCConnect returns −1	Host unreachable or routing timeout.	Run ping <ip> and Test-NetConnection -ComputerName <ip> -Port <port> to check the network path.
Connects but CONNECTED NO	Wrong IP/credentials, or OPC UA disabled on the CNC.	Verify SIEMENS_USER / SIEMENS_PASS and endpoint address; confirm the OPC UA server is enabled on the NCU.
BadRequestNotAllowed	Siemens certificate trust rejection.	Export client_cert.der into the NCU certificate storage under trusted certificates.
BadNodeIdUnknown	OPC UA node is missing on the target model.	Node is unavailable for this hardware model; the field will default to 0. Use check mode (go run sinumerik_client.go

Observable Error / Symptom	Root Cause Analysis	Remediation Step
		check) to enumerate available nodes.
Garbled console output	Old console window has a broken terminal mode.	Close and reopen the terminal window.